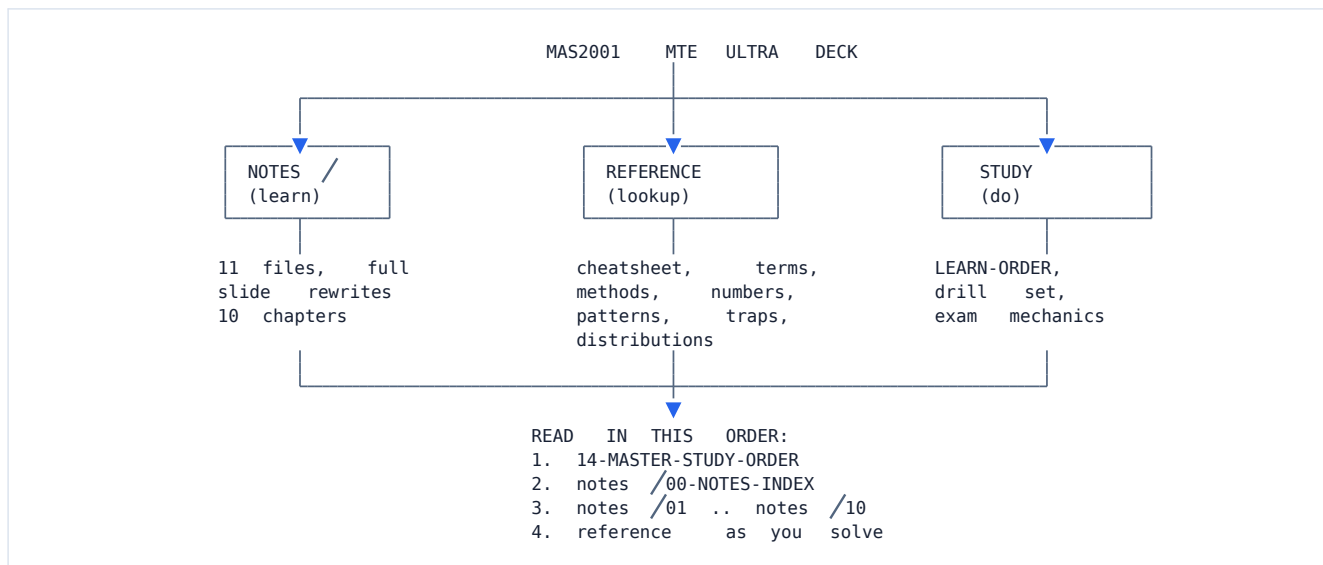
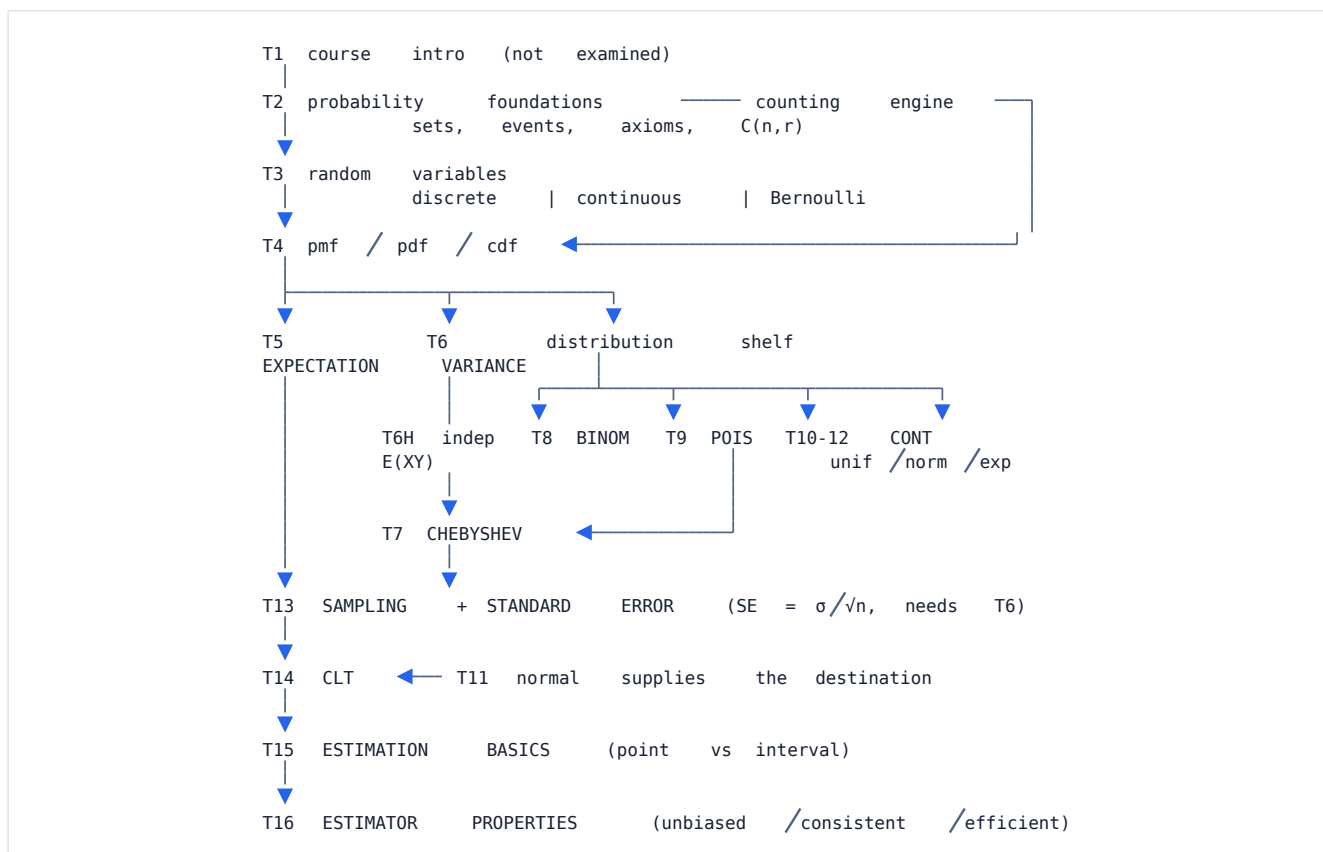


00 INDEX: the map

THE WHOLE FOLDER AS A MAP



THE COURSE SPINE, TOP TO BOTTOM



THE HIDDEN LAYER, ON THE SPINE

the nodes the slides UNDER-TAUGHT but the papers ASK:

T6 —[H1 independence rules] → T7, T14, T16 <- gates 3 topics
 T7 —[H2 Chebyshev] → graded 9+ times
 T12 —[H3 memoryless] → A2 MCQ5, A2 C1
 T11 —[H4 68 /95 /99.7] → A2 MCQ11
 T5 —[H5 N x P count] → A1 app, A2 B4, A2 D1
 T3 —[H6 geometric 1/p] → A1 short4
 T3 —[H7 hypergeometric nK /N] → A1 long3
 T11 —[H8 two table conventions] → every normal /CLT numeric

= the cheapest marks on the paper. See notes /10 + 11-BEYOND-SLIDES.

1. What the MTE is

paper	MAS2001 Statistics and Probability, mid term
marks	30
scope	lectures 1 to 21 (the 16 syllabus lines below)
shape	Section A MCQ (a few marks each), Section B short answers,
Section C longer. All compulsory.	
closed book so the sheet below must be in your head	
calculator is allowed on both supplied MTE papers	

2. The 16 syllabus lines and where each is taught

#	syllabus line	lecture	file
1	Introduction of the Course	1	md/notes-lecture-series-01-09 p001-011
2	Basic terminology and concepts of probability	2	notes p012-050
3	Random variables (discrete, continuous)	3-4	notes p051-062, p112
4	PMF, pdf, CDF	5-6	notes p063-087, p112-139
5	Expectation of random variables	7	notes p088-094, p140-147
6	Expectation and Independent random variables	8-9	notes p095-111 (indep rules: hidden, 11)
7	Chebyshev's inequality	10-11	deck L10-11 (9 pages)
8	Binomial distribution	12	md/ppt3 p001-018
9	Poisson distribution	13	md/ppt3 p019-028
10	Uniform distribution (continuous)	14	md/ppt4 p001-006
11	Normal distribution	15	md/ppt4 p007-037
12	Exponential distribution	16	md/ppt4 p038-043
13	Sampling concepts, standard error	17	md/lms-standard-error-clt p001-019
14	Central limit theorem	18	md/lms-standard-error-clt p006-018
15	Theory of estimation: parameter, statistic,		
point and interval estimation	19	md/ppt5 p001-007, lms-theory p001-015	
16	Characteristics of a good estimator	20-21	md/ppt5 p008-021, lms-theory p016-030

3. The 16-topic tree (full detail in 05 and 10)

T1	course intro (not examined)
T2	probability foundations: sets, events, axioms, counting, conditional, independence
T3	random variables: definition, discrete, continuous, functions of an rv
T4	pmf, pdf, cdf: conditions, build, read, invert
T5	expectation: definition, $E[h(X)]$, linearity, expected count
T6	variance: definition, shortcut, transforms; independent rvs ($E(XY)$, Var of sum)
T7	Chebyshev: tail form, complement form, k extraction, apply, inverse
T8	binomial
T9	Poisson
T10 uniform	
T11 normal	
T12 exponential	
T13 sampling and standard error	
T14 central limit theorem	
T15 estimation basics	
T16 estimator properties	

4. The learning order (why, and the evidence)

The batch-2 S&P decks are re-teaches, proved by shared-phrase counts:

L1-7	vs notes deck	1,129 shared 5-grams (54 percent of the smaller)
L8-9	vs notes deck	456 shared (55 percent)
L12-13 vs ppt3	874 shared (53 percent), same pens numbers 0.2301, 0.2824, 0.3766	
L14-15 vs ppt4	630 shared (40 percent)	
L10-11 vs notes deck	12 shared (2 percent)	<- the only genuinely new deck

So the order is: Chebyshev and the hidden layer first (nothing else teaches them, they gate four downstream topics), then the main theory, with each S&P re-teach used as the second pass on its topic, then the papers, then the assignments.

PASS 0	Chebyshev deck + independence rules + the hidden set + formula sheet	1h40
PASS 1	notes p012-087 (foundations, rvs, pmf / cdf) + its questions	3h
PASS 2	notes p088-147 (expectation, variance) + S&P L1-7 and L8-9 questions	3h15
PASS 3	binomial, Poisson, uniform, normal, exponential + their re-teach blocks	4h
PASS 4	sampling and CLT + questions	2h
PASS 5	estimation (ppt5 full, lms-theory 2 items only) + questions	2h30
PASS 6	mock paper, closed book, timed, then mark it	3h
PASS 7	MTE papers, then assignments, then ETE IN blocks	several days
PASS 8	sheet, landmarks, hidden set, traps, from memory	last day

Full per-question sequencing: [../reports/11-QUESTION-ATLAS/00-LEARN-ORDER.md](https://reports/11-QUESTION-ATLAS/00-LEARN-ORDER.md).

5. The counts you can trust (as of 16 Sep 2026)

base branch ledger	383 instances	
assignment-bundle addition	119 instances on PR 14	
combined enumerated total	502 instances after PR 14 is merged	
MTE source/family leads	10 of 16 blocks	
MTE generic concept checks	5 of 16 blocks	
MTE open composite	1 of 16 blocks	
content family ids	0 of 502 rows, dedup remains unfinished	
exam papers on disk	9	(2 MTE + 4 ETE + 2 re-sess + 1 summer)
marking schemes on disk	2	(kept separate from exam-paper count)

The 16 MTE blocks have all been searched. “Searched” is not the same as “traced”. A generic definition match is not evidence that a question was copied from a particular source.

6. The hidden layer, one line each (full detail in 11)

H1	independence rules of rvs: $E(XY)$, $\text{Var}(X+Y)$, $\text{Var}(X-Y)$	no teaching slide
H2	Chebyshev entire topic	taught only in the L10-11 deck
H3	memoryless property of the exponential	no slide
H4	the 68.27 / 95.45 / 99.73 landmarks	no slide
H5	expected count, $N \times P$	no slide
H6	geometric mean $1/p$	no slide
H7	hypergeometric mean nK/N	no slide
H8	two normal table conventions (ϕ vs F)	no slide